



Technical Service Bulletin: TSB110126

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Injector Set Frequency

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Note : Remove topic 93T6-18 and replace with this topic. The Fixed Time(s) in Table 1 have been corrected.

This topic is being issued to highlight the recent change in the recommended VALVE AND INJECTOR SET AND RESET INTERVALS for all K engines. This recommendation became effective July 1991 when issued in the K19 Operation and Maintenance Manual, Bulletin 3666013-00, and the K38 and K50 Operation and Maintenance Manual, Bulletin No. 3810497-01. These recommendations continue in the July and August 1993 update of these Operation and Maintenance Manuals (K19, Bulletin No. 3666013-01; K38 and K50, Bulletin No. 3810491-02).

These manuals state:

Cummins® K engines in most applications will **not** experience significant valve and injector train wear after an initial adjustment is made at 1,500 hours. (Reference Table 1 for the injector set methods and values per injector.) After this adjustment, Cummins® recommends the valves and injectors **not** be adjusted again until the 6,000 hours or 2 year injector calibration interval. Because injector train hardware is typically mixed between cylinders during injector replacement, Cummins® recommends to adjust valves and injectors at 1,500 hours after all injector replacements.

If the valve or injector adjustment is checked during troubleshooting or before the recommended maintenance interval, adjustment is **not** required if measurements are within the recheck limits. (Reference Table 1.)

Simplified Injector Setting Recommendations

1. Install the injectors, then adjust the crossheads, valves and injectors.
2. Operate the engine 1,500 hours and re-adjust the valves and injectors.
3. Operate the engine an additional 4,500 hours (6,000 hours on injectors), then clean and calibrate or replace the injectors. Start with Step 1 again.

Fixed time engine injectors are set by adjusting the travel using a dial indicator. This is the **only** acceptable method for adjusting these injectors.

STV and HVT injectors **must** be adjusted using the Outer Base Circle (OBC) method. Use this method rather than previous methods because:

1. It is much easier
2. There is less chance of errors
3. STC injectors do **not** have to be removed to set the total travel.

The injector set methods and values for the different injectors are specified below.

Table 1: Crosshead, Valve and Injector Settings and Recheck Limits		
Crosshead, Valve, Injector Type	Normal Setting	Reset Limits
Fixed Time (K38 and K50)	7.811 mm [0.3075 in] Travel	7.75 to 7.87 mm [0.305 to 0.310 in] Travel
Fixed Time (K19)	7.72 mm [0.304 in] Travel	7.67 to 7.77 mm [0.302 to 0.306 in] Travel
STC and HVT (OBC)	10 N•m [89 in-lb]	10 N•m [89 in-lb]
Crosshead Lash	Zero	Zero
Intake Valve Lash	0.0356 mm [0.014 in]	0.279 to 0.432 mm [0.011 to 0.017 in]
Exhaust Valve Lash	0.686 mm [0.027 in]	0.610 to 0.763 mm [0.024 to 0.030 in]

Injector adjustment sequence charts for the OBC adjustment method are included in this topic. These charts can also be found in the Operation and Maintenance Manuals listed above, Service Parts Topic 92T6-2, and in Bulletin 3666049 (a plasticized card).

If problems of injector or valve train wear occur, the previous recommendation of setting valves and injectors every 1,500 hours is to be used.

Note : Two full engine rotations are required to adjust the valves and injectors on K19, K38, and K50 engines.

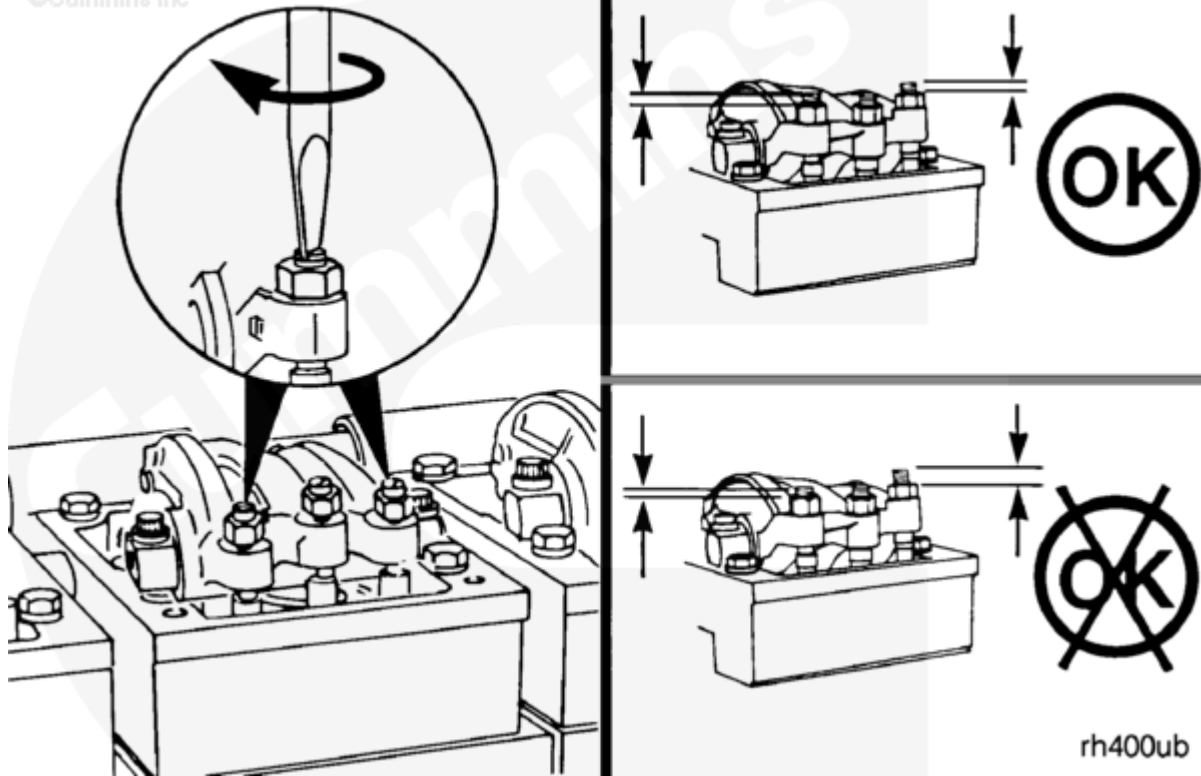
Determine the Cylinder in Position for Valve Set

The crosshead and valves can be adjusted on a cylinder that has all of the valves closed. Reference the tables on the following pages.

The same procedure can be used to determine the cylinder in position for valve setting for both methods (travel method or OBC method). However, different injectors will be adjusted. If the rocker lever assemblies have been removed, use this step to determine the cylinder to set.

Note : All adjusting screws must be loose on all cylinders, and the push rod must remain in alignment.

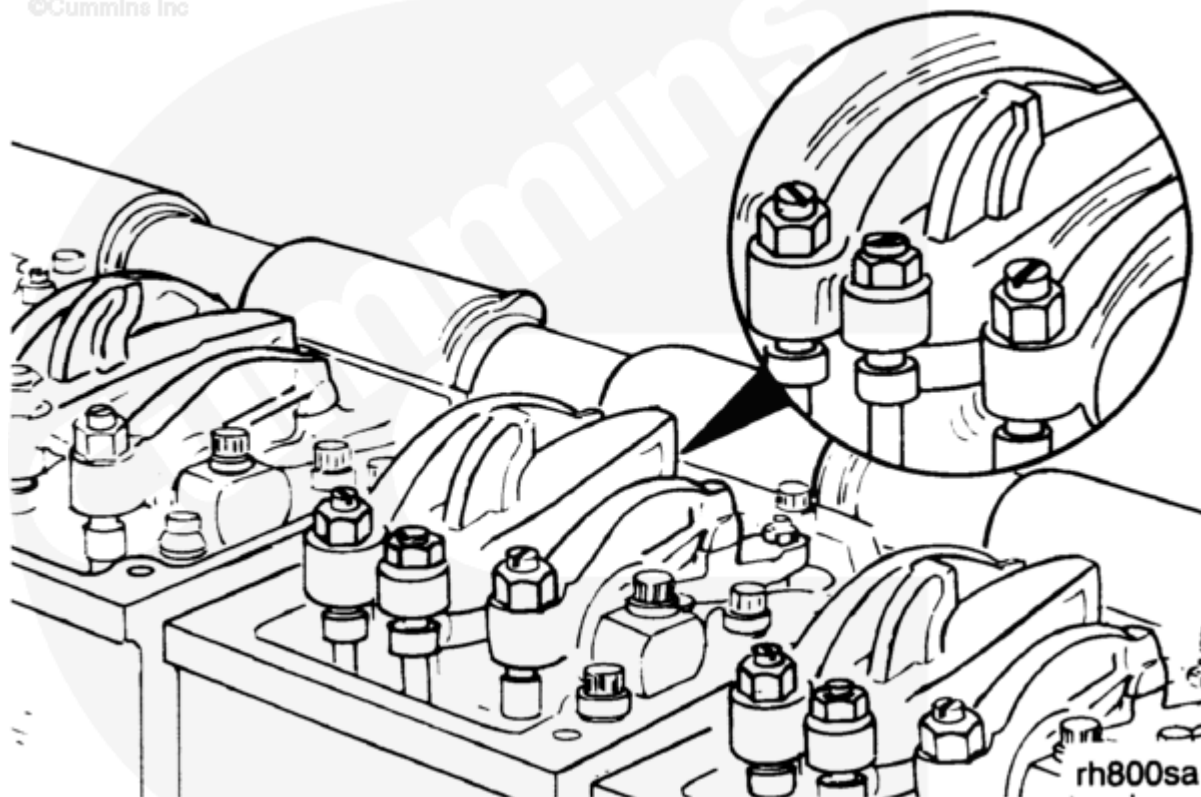
Perform this step on both cylinders to be checked.



Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the locknuts until they touch the levers.

Compare the height of the adjusting screws above the locknut. The cylinder with the adjusting screws that are nearest to the same height has both valves closed.

The push rods will be close to the same height above the top of the rocker lever housing on the cylinder that has both valves closed.



Note : One engine mode, KTTA19-G/GSGC2, CPL1170, contains a unique camshaft that creates a noticeable difference in the height of the valve adjusting screws. When the valves are properly adjusted on CPL1170 **only**, the exhaust valve adjusting screw will have approximately one thread visible **above** the top of the locknut. The intake valve adjusting screw will have approximately 3 threads visible **above** the top of the adjusting screw.

If the rocker levers have **not** been removed, wiggle the valve rocker levers on the two cylinders in question. The crossheads and valves on the cylinder where both levers feel loose are ready to adjust.

Use the chart to determine the injector that is ready to adjust.

Note : Adjustment can begin on any valve set mark.

TRAVEL METHOD INJECTOR AND VALVE SETTING PROCEDURE FOR FIXED TIME PT (type D) INJECTORS

Note : The Check Valve Position On Cylinder portion of this K19 chart has been changed from the previous publications. The change was made for clarification of which cylinder has valves open or closed so engine rotational position can be determined more clearly. The valves and injectors to be adjusted at a given mark (rotational position) has **not** changed. Valve and injector adjustment sequence has **not** changed.

K19 Fixed Time PT (Type D) Valve and Injector Adjustment

VS Mark	Valves Closed On	Set Valves On	Set Injector On
A	1	5	3
B	5	3	6
C	3	6	2

K19 Fixed Time PT (Type D) Valve and Injector Adjustment			
VS Mark	Valves Closed On	Set Valves On	Set Injector On
A	6	2	4
B	2	4	1
C	4	1	5

K38 Fixed Time PT (Type D) Valve And Injector Adjustment Chart		
VS Mark	Valves Closed On Cylinder Number	Adjust Valves And Injectors On Cylinder Number
1R-6R VS	1 RB	1 RB
6L-1L VS	6 LB	6LB
5R-2R VS	5RB	5 RB
2L-5L VS	2 LB	2 LB
3R-4R VS	3 RB	3 RB
4L-3L VS	4 LB	4 LB
1R-6R VS	6 RB	6 RB
6L-1L VS	1 LB	1 LB
5R-2R VS	2 RB	2 RB
2L-5L VS	5 LB	5 LB
3R-4R VS	4 RB	4 RB
4L-3L VS	3 LB	3 LB

K50 Fixed Time PT (Type D) Valve And Injector Adjustment Chart (Fixed Time Injectors)		
VS Mark	Valves Closed On Cylinder Number	Adjust Valves And Injectors On Cylinder Number
1R-8R VS	1 RB	1 RB
1L-8L VS	1 LB	1 LB
3R-6R VS	3 RB	3 RB
3L-6L VS	3 LB	3 LB
2R-7R VS	7 RB	7 RB
2L-7L VS	7 LB	7 LB
4R-5R VS	5 RB	5 RB
4L-5L VS	5 LB	5 LB
1R-8R VS	8 RB	8 RB
1L-8L VS	8 LB	8 LB
3R-6R VS	6 RB	6 RB

K50 Fixed Time PT (Type D) Valve And Injector Adjustment Chart (Fixed Time Injectors)

VS Mark	Valves Closed On Cylinder Number	Adjust Valves And Injectors On Cylinder Number
3L-6L VS	6 LB	6 LB
2R-7R VS	2 RB	2 RB
2L-7L VS	2 LB	2 LB
4R-5R VS	4 RB	4 RB
4L-5L VS	4 LB	4 LB

OUTER BASE CIRCLE (OBC) INJECTOR AND VALVE SETTING PROCEDURE FOR K SERIES ENGINES

These charts **must** be used when using the OBC Injector Set Method on K engines that have STC and HVT injectors. Details of the OBC set procedure are specified in SPT 92T6-2. The setting specifications are:

Injectors:	10 N•m [89 in-lb]
Intake Valve Lash:	0.036 mm [0.014 in]
Exhaust Valve Lash:	0.069 mm [0.027 in]

Note : When setting the injectors, the rocker lever adjusting screw **must** to be torqued to 11 N•m [97 in-lb], loosen the adjusting screw, then torque the adjusting screw again to the 10 N•m [89 in-lb] specification.

PT (Type D) fixed time injectors **must not** to be adjusted using the OBC method. Fixed time PT (Type D) injectors **must** be adjusted using travel method.

Note : The Check Valve Position On Cylinder portion of this K19 chart has been changed from the previous publications. The change was made for clarification on which cylinder has valves open or closed so engine rotational position can be determined more clearly. The valves and injectors to be adjusted at a given mark (rotational position) has not changed. Valve and injector adjustment sequence has not changed.

K19 OBC Set Procedure With STC or HVT and Injector Adjustment

VS Mark	Valves Closed On Cylinder Number	Adjust Valves On Cylinder Number	Adjust Injectors On Cylinder Number
A	1	5	4
B	5	3	1
C	3	6	5
A	6	2	3
B	2	4	6
C	4	1	2

K38 OBC Set Procedure With STC Or HVT Valve And Injector Adjustment Chart

VS Mark	Valves Closed On Cylinder Number	Adjust Valves On Cylinder Number	Adjust Injectors On Cylinder Number
1R-6R VS	1R	1R	2R
6L-1L VS	6L	6L	5L
5R-2R VS	5R	5R	4R
2L-5L VS	2L	2L	3L
3R-4R VS	3R	3R	1R
4L-3L VS	4L	4L	6L
1R-6R VS	6R	6R	5R
6L-1L VS	1L	1L	2L
5R-2R VS	2R	2R	3R
2L-5L VS	5L	5L	4L
3R-4R VS	4R	4R	6R
4L-3L VS	3L	3L	1L

⚠ CAUTION ⚠

For K50 STC or HVT engines, it is important to know if the engine has the Standard Firing Order or the REVISED FIRING ORDER. Do NOT use the Standard Firing Order sequence for update engines manufactured after September 1986 that have REVISED FIRING ORDER. All engines that have the REVISED FIRING ORDER have STC or HVT injectors and are identified as Revised on the engine dataplate. These engines also have decals on the rocker lever covers.

K50 OBC Set Procedure - Revised Firing Order With STC Or HVT Valve And Injector Adjustment Chart

VS Mark	Valves Closed On Cylinder Number	Adjust Valves On Cylinder Number	Adjust Injectors On Cylinder Number
1R-8R VS	1R	1R	6R
1L-8L VS	1L	1L	6L
3R-6R VS	3R	3R	7R
3L-6L VS	3L	3L	7L
2R-7R VS	2R	2R	4R
2L-7L VS	2L	2L	5L
4R-5R VS	5R	5R	1R
4L-5L VS	4L	4L	1L
1R-8R VS	8R	8R	3R

**K50 OBC Set Procedure - Revised Firing Order With STC Or HVT Valve And Injector
Adjustment Chart**

VS Mark	Valves Closed On Cylinder Number	Adjust Valves On Cylinder Number	Adjust Injectors On Cylinder Number
1L-8L VS	8L	8L	3L
3R-6R VS	6R	6R	2R
3L-6L VS	6L	6L	2L
2R-7R VS	7R	7R	5R
2L-7L VS	7L	7L	4L
4R-5R VS	4R	4R	8R
4L-5L VS	5L	5L	8L

**K50 OBC Set Procedure - Standard Firing Order With STC Or HVT Valve And Injector
Adjustment Chart**

VS Mark	Valves Closed On Cylinder Number	Adjust Valves On Cylinder Number	Adjust Injectors On Cylinder Number
1R-8R VS	1R	1R	6R
1L-8L VS	1L	1L	6L
3R-6R VS	3R	3R	2R
3L-6L VS	3L	3L	2L
2R-7R VS	7R	7R	4R
2L-7L VS	7L	7L	4L
4R-5R VS	5R	5R	1R
4L-5L VS	5L	5L	1L
1R-8R VS	8R	8R	3R
1L-8L VS	8L	8L	3L
3R-6R VS	6R	6R	7R
3L-6L VS	6L	6L	7L
2R-7R VS	2R	2R	5R
2L-7L VS	2L	2L	5L
4R-5R VS	4R	4R	8R
4L-5L VS	4L	4L	8L

Document History

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